



TEIRAC PRIVATE LIMITED

Architecting Resilience: Strategic Tech Solutions for
Enterprise, Energy, and Education

Anupam
Co-Founder & COO

OPTOPROFIT ENGINE

Optimization of production processes to enhance profitability via improved throughput, waste reduction, and efficiency.

Assembly Line Optimization for Digital Oscilloscopes

Background: Company XYZ Pvt Ltd, an electronics manufacturer produces digital oscilloscopes. The current manufacturing process involves eight discrete operational tasks, ranging from initial PCB preparation to final QA and packaging. The total processing time required to complete a single unit is 130 minutes. The facility currently operates on a single 8-hour (480-minute) daily shift.

The Core Challenge

The manufacturer has no balanced, math-based workflow and uses poor task order. This causes idle time, production jams, excess half-made inventory, and higher costs. To hit 16 units daily and stay profitable without expanding too soon, it needs a streamlined, single-line "cellular" layout.

Key Objectives

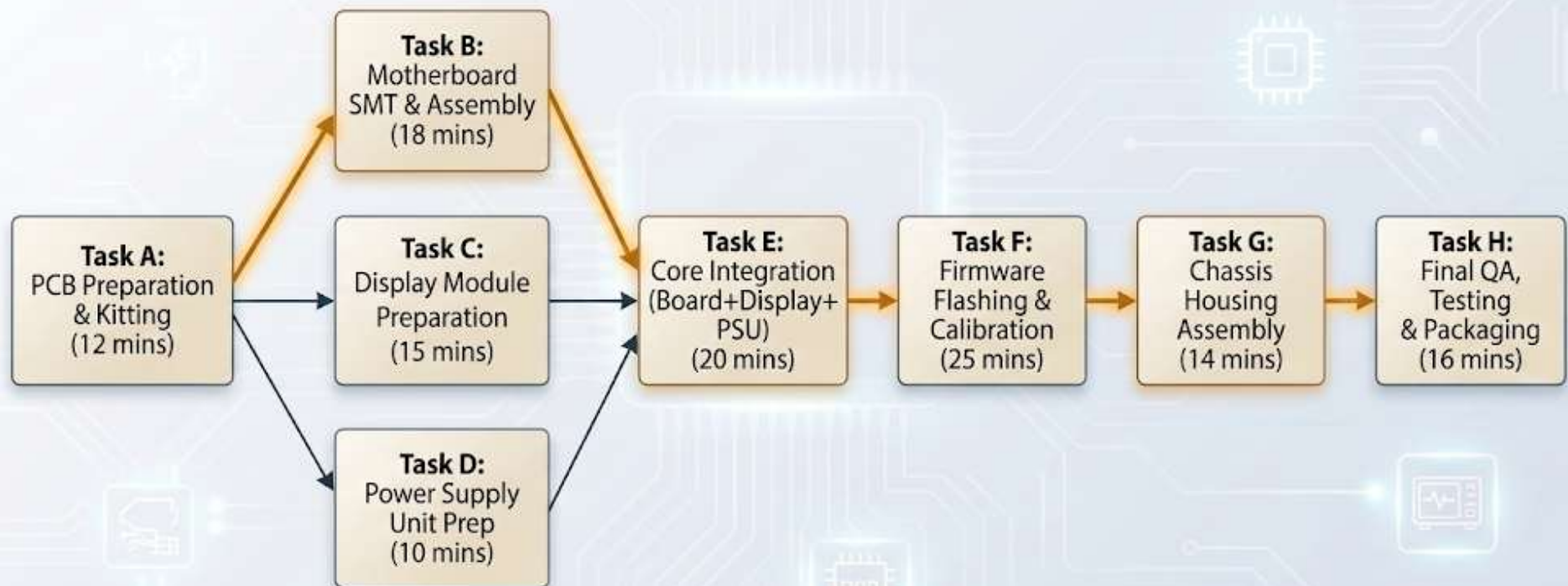
1. **Establish Cycle Time:** Figure out the exact speed needed to make 16 units in a 480-minute shift.
2. **Define Spatial Requirements:** Find out the smallest number of different workstations needed on the factory floor, in theory.
3. **Optimize Task Allocation:** Group the assembly tasks into workstations using the longest-task-first method, while strictly following the task order rules (e.g., Task D can't start until A, B, and C are done).
4. **Maximize Resource Yield:** Find the main production jam and reach at least 85% line efficiency to cut down on wasted worker time.

Break down the assembly of an oscilloscope into discrete, measurable tasks, establish their standard times, and identify their immediate predecessors.

Task	Description	Immediate Predecessor(s)	Task Time (t) in mins
A	PCB Preparation & Kitting	None	12
B	Motherboard SMT & Assembly	A	18
C	Display Module Preparation	A	15
D	Power Supply Unit (PSU) Prep	A	10
E	Core Integration (Board + Display + PSU)	B, C, D	20
F	Firmware Flashing & Calibration	E	25
G	Chassis Housing Assembly	F	14
H	Final QA, Testing & Packaging	G	16
	Total Task Time ($\sum t$)		130 mins

PRECEDENCE NETWORK DIAGRAM

OSCILLOSCOPE ASSEMBLY LINE (TOTAL 130 MINS)



The Mathematical Output: Balancing the Line

Company XYZ Pvt Ltd has a target demand of 16 oscilloscopes per day.

Step 1: Calculate Cycle Time (C)

$$C = \frac{\text{Available Production Time}}{\text{Daily Demand}} = \frac{480 \text{ mins}}{16 \text{ units}} = 30 \text{ mins/unit}$$

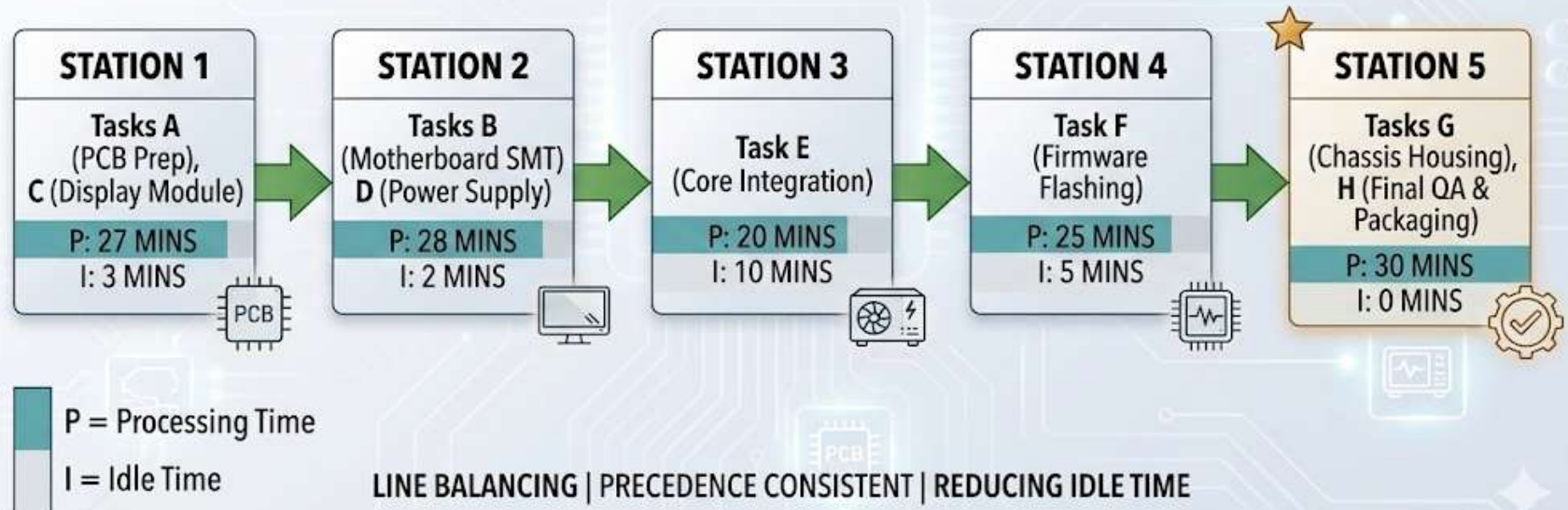
The Mathematical Output: Balancing the Line

Step 2: Calculate Theoretical Minimum Workstations (N_{min})

$$N_{min} = \frac{\sum t}{C} = \frac{130}{30} = 4.33$$

MANUFACTURING CELL LAYOUT DIAGRAM: OSCILLOSCOPE ASSEMBLY LINE

DAILY DEMAND: 16 UNITS | CYCLE TIME (C): 30 MINS | LINE EFFICIENCY: 86.67% | THEORETICAL WORKSTATIONS (N_{MIN}): 5



Step 3: Task Allocation (Line Balancing)

Using the "longest task first" rule while following the task order diagram, group tasks into 5 workstations so none exceeds the 30-minute cycle time.

Workstation	Assigned Tasks	Processing Time	Idle Time (C–Processing Time)
Station 1	A (12), C (15)	27 mins	3 mins
Station 2	B (18), D (10)	28 mins	2 mins
Station 3	E (20)	20 mins	10 mins
Station 4	F (25)	25 mins	5 mins

Step 4: Performance Metrics

- i. Actual Workstations (N): 5
- ii. Total Idle Time: 20 minutes per cycle
- iii. Line Efficiency (E):

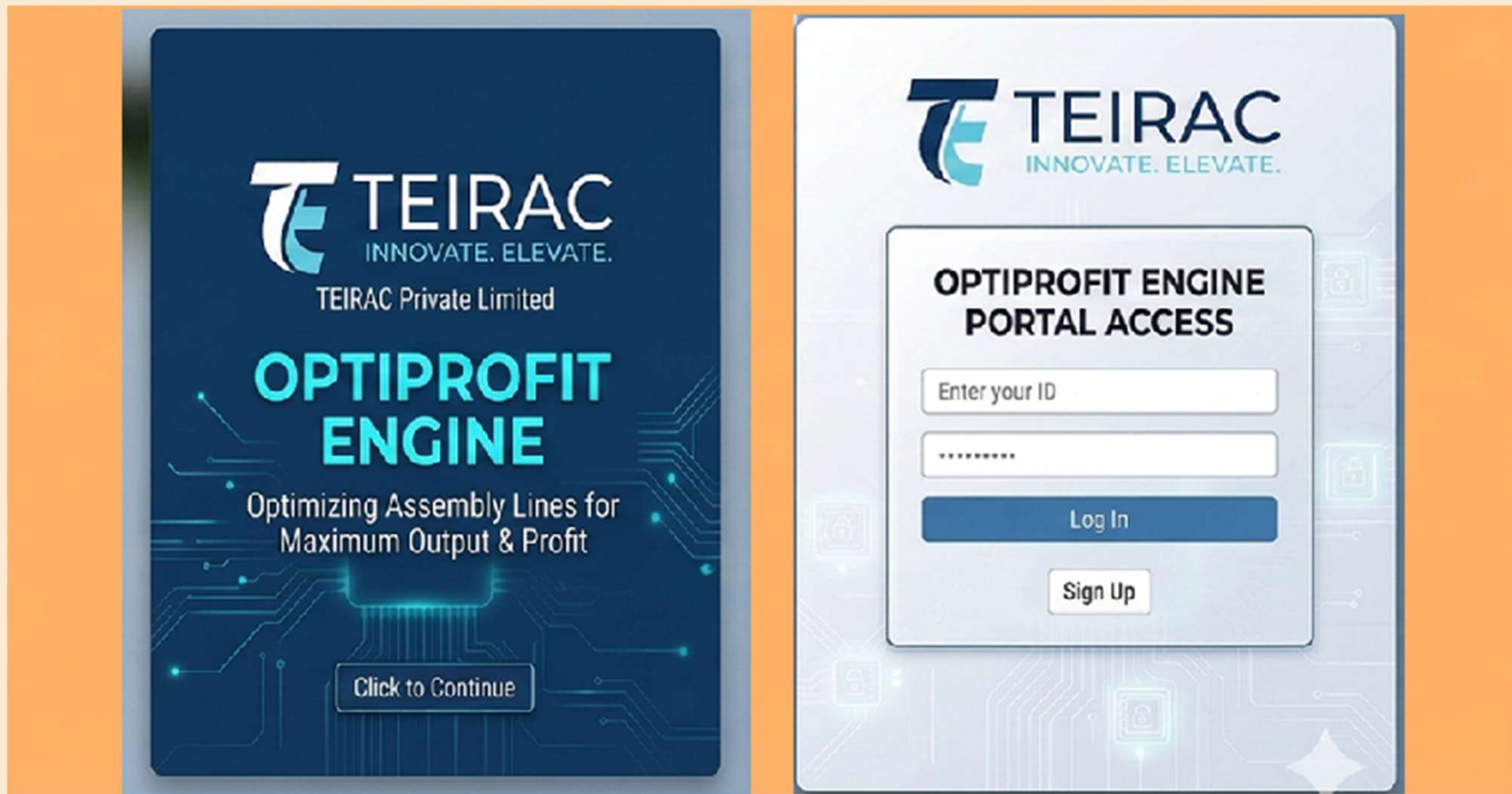
$$E = \frac{\sum t}{N \cdot C} \times 100\% = \frac{130}{5 \cdot 30} \times 100\% = \frac{130}{150} \times 100\% = 86.67\%$$

OPTOPROFIT ENGINE

APPLICATION REQUIREMENTS

1. Welcome Screen (branding, login)
2. Dashboard (analytics overview)
3. Process Planning (data input forms)
4. Precedence Network Diagram (task dependencies visualization)
5. Conceptual Layout Diagram (initial flow)
6. Line Optimization (algorithms for balancing)
7. Floor Layout Diagram (2D/3D rendering)
8. Financial Performance Matrices (ROI calculators)

1. WELCOME SCREEN



2. DASHBOARD



3. PROCESS PLANNING



PROCESS PLANNING | Task & Precedence Data Entry

Manufacturing Line: Oscilloscope Model-X.

Save Process
Export Data
Visualize Flow

Task ID	Task Description	Standard Time (mins)	Immediate Predecessors
A	PCB Preparation & Kitting	12	None
B	Motherboard SMT & Assembly	18	A
C	Display Module Preparation	15	A
D	Power Supply Unit (PSU) Prep	10	A
E	Core Integration (Board + Display + PSU)	20	B, C, D
F	Firmware Flashing & Calibration	25	E
G	Housing Assembly	14	H, A
H	Housing Assembly		

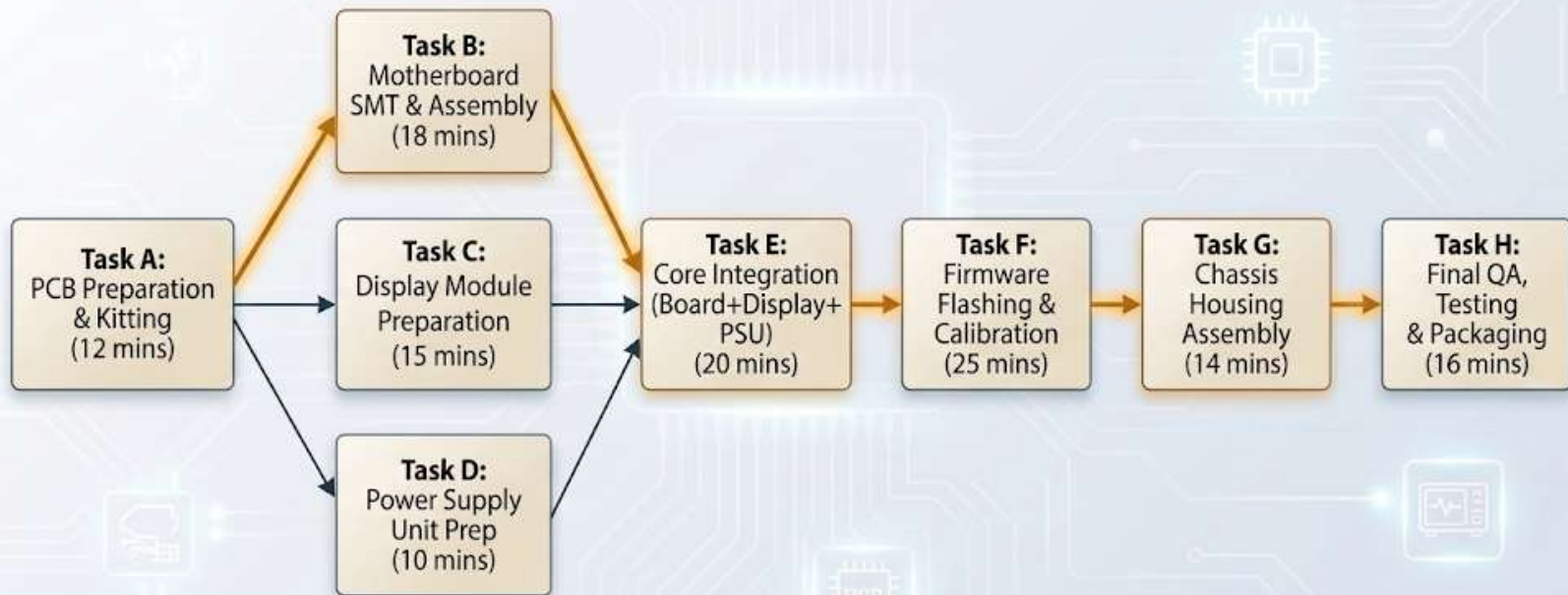
+ **Add Task**

Press Enter to save row. Circles will be auto-validated.

Example Predecessors: A, B, C (separate with commas)

4. PRECEDENCE NETWORK DIAGRAM

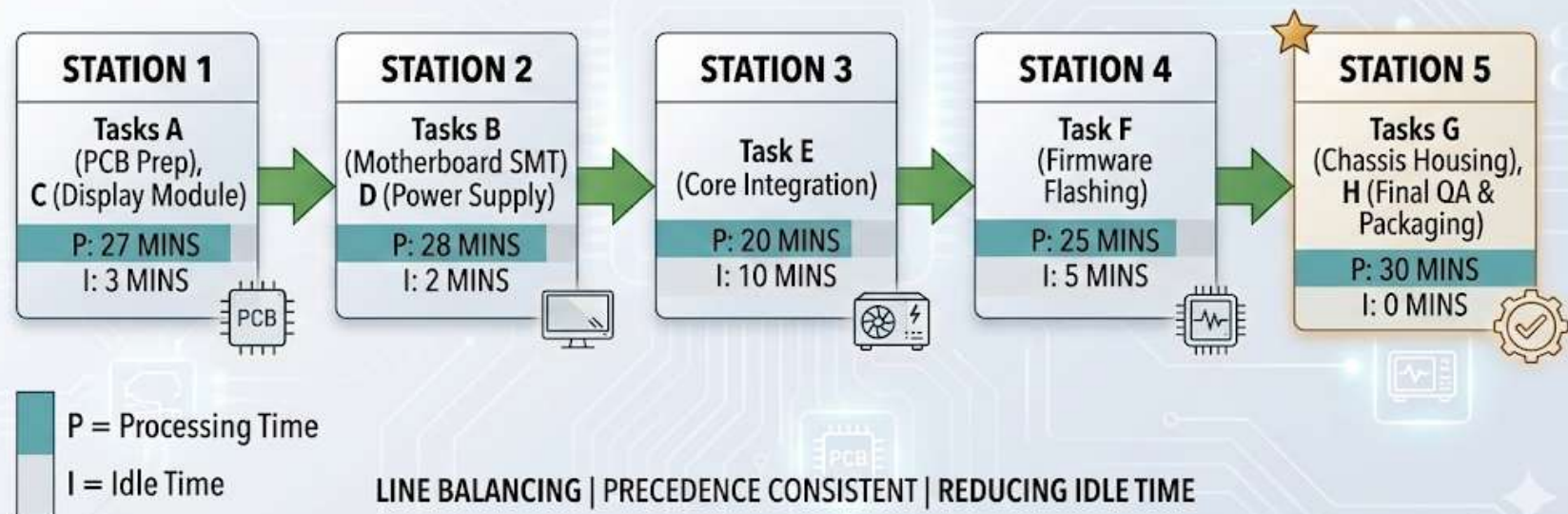
PRECEDENCE NETWORK DIAGRAM OSCILLOSCOPE ASSEMBLY LINE (TOTAL 130 MINS)



5. CONCEPTUAL LAYOUT DIAGRAM

MANUFACTURING CELL LAYOUT DIAGRAM: OSCILLOSCOPE ASSEMBLY LINE

DAILY DEMAND: 16 UNITS | CYCLE TIME (C): 30 MINS | LINE EFFICIENCY: 86.67% | THEORETICAL WORKSTATIONS (N_{MIN}): 5



6. LINE OPTIMIZATION

INDUSTRIAL ENGINEERING TOOLKIT - LINE BALANCING ENGINE

FileEditLine PlanOptimizationReports

☰

🕒

📊

⚙️

Optimization Controls

Heuristic Selection

Longest Task Time

Longest Task Time

Most Following Tasks

Target Cycle Time:

30 seconds

(optional)

Number of Operators:

4

(optional)

Task ID	Task Description	Time (seconds)
A	Prep	12s
B	Assemble	15s
C	Connect	15s
D	Test	27s
E	Pack	30s
F	Assemble	12s
G	Connect	15s
H	Test	27s

OPTIMIZE LINE

Optimized Line Layout

Line Efficiency:

86.7%

Total Idle Time (per cycle):

23 seconds

STATION 1

A (Prep)

12s

C (Connect)

15s

Total: 27s

STATION 2

B (Prep)

2s

E (Connect)

17s

Total: 32s

STATION 3

D (Prep)

12s

F (Test)

15s

Total: 33s

STATION 4

G (Prep)

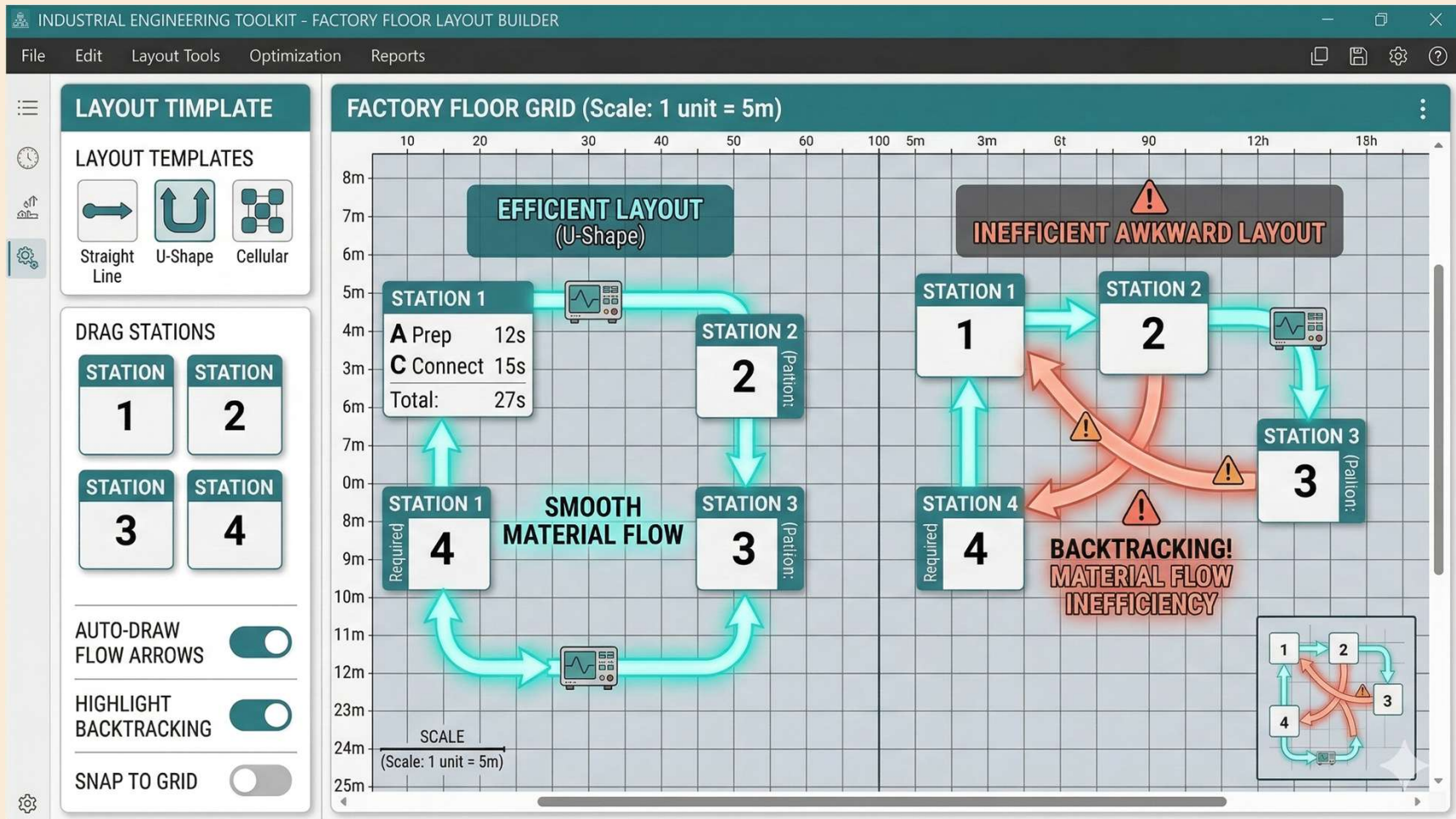
2s

H (Pack)

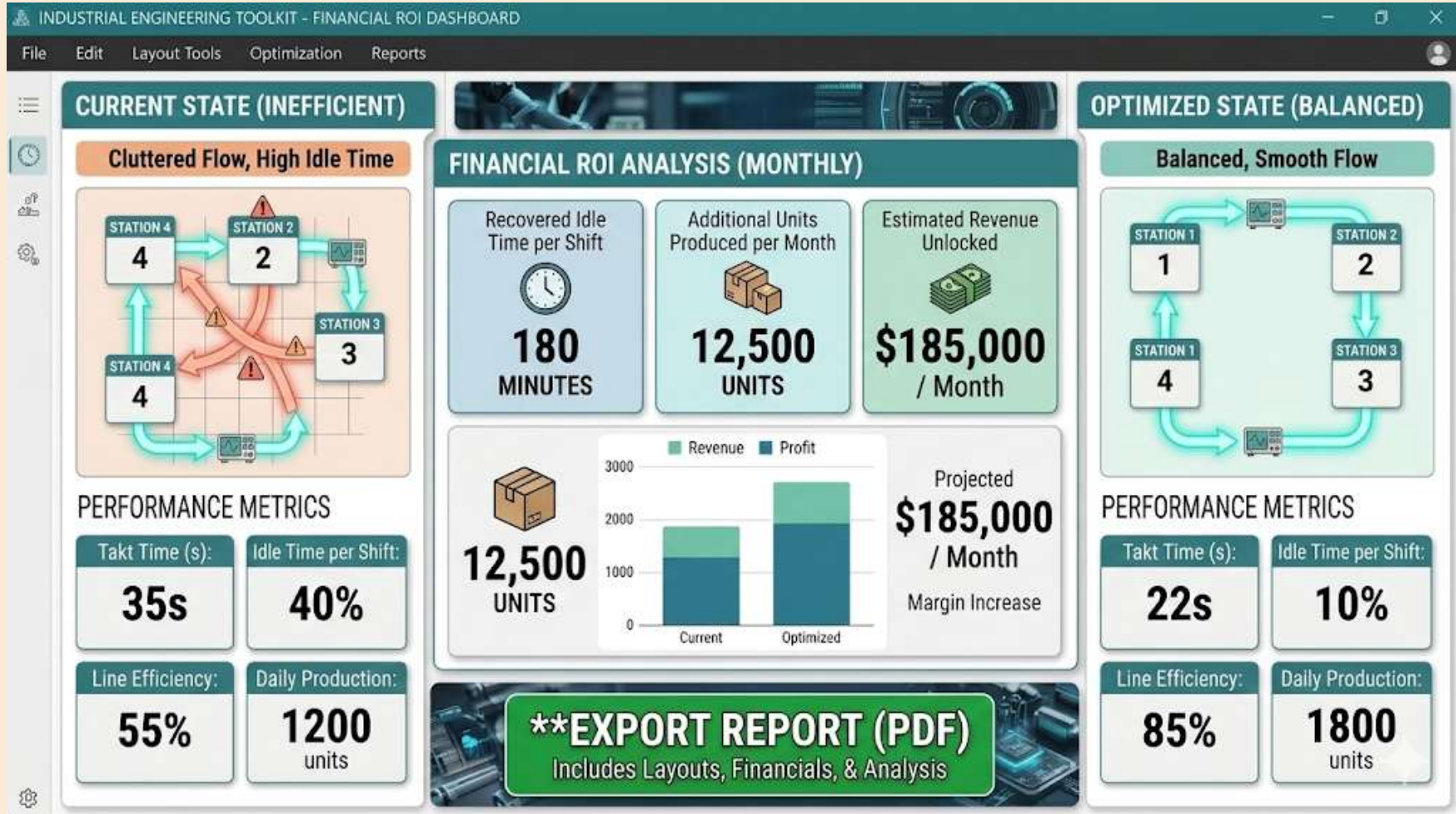
10s

Total: 30s

7. FLOOR LAYOUT DIAGRAM



8. FINANCIAL PERFORMANCE MATRICES



Thank You